Maths Test

Section A

1. 3

2. $\tan 60^\circ$

3. 30°

4. $\frac{20}{3}$ cm

5. $a = 0$

6. Two decimal places

7. 3 and 5

8. $(0, -1)$

9. Infiniatly many solutions

10. $k = \frac{-9}{2}$

11. $k \neq -11$

12. 6

3. Both (1) and (2)

4. $\cancel{x^5y^5z^6}$ $x^5y^5z^6$

$\frac{x^2}{2} - \frac{x}{2} - 6$

$$\frac{(\sin^2\theta + \cos^2\theta + \cos\theta\sin\theta)(\sin\theta - \cos\theta)}{\sin\theta\cos\theta[\sin\theta - \cos\theta]}$$

$$\frac{1 + \cos\theta\sin\theta}{\sin\theta\cos\theta}$$

$$\frac{1}{\sin\theta\cos\theta} + 1$$

$$1 + \sec\theta\,\mathrm{cosec}\,\theta$$

hence, proved.

33. ~~Factorisation~~ -

$x - y + 1 = 0$

$x =$

Section - E

36. (i) ~~(12, 0)~~ (0, 12)

(ii) (3, 5)

(iii) 5m

(iii) [illegible]

$\cancel{16x = 164}$

$x^2 + 25 - 10x + 4 = x^2 + 9 + 6x + 4$

$x^2 - x^2 - 10x - 6x = 9 - 25 - 4 + 4$

$-16x = 9 - 25 - 0$

$\cancel{-}16x = \cancel{-}16$

$\frac{16x}{16} = x$

$1 = x$

$\boxed{x = 1}$

Hence, The point is 1

Section-D

32.

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\,\text{cosec}\,\theta$$

~~Divide by $\sin\theta$. [L.H.S]~~

$\Rightarrow$ $\cancel{\frac{\sin\theta}{\cos\theta} + \cot\theta}$

$\cancel{\frac{\tan\theta}{\cos\theta}}$ $\quad \dfrac{\dfrac{\sin\theta}{\cos\theta}}{\dfrac{\sin\theta-\cos\theta}{\sin\theta}} + \dfrac{\dfrac{\cos\theta}{\sin\theta}}{\dfrac{\cos\theta-\sin\theta}{\cos\theta}}$

$$\frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} + \frac{\cos^2\theta}{\sin\theta(\cos\theta-\sin\theta)}$$

$$\frac{\sin^2\theta}{\cos\theta[\sin\theta-\cos\theta} - \frac{\cos^2\theta}{\sin\theta[\sin\theta-\cos\theta)}$$

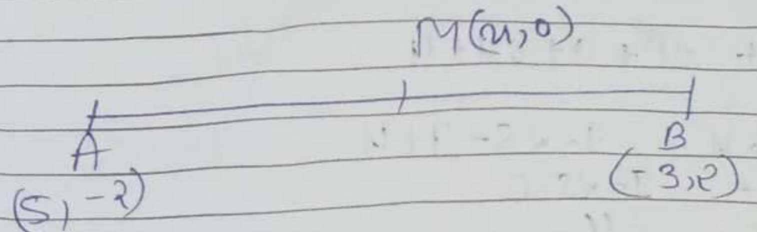
$$\frac{\sin^3\theta-\cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)}$$

144
20
164

−20
144

30.

31.



Let the point on 'X' axis be M(x,0)

<u>Given</u> $x_1 = x \quad y_1 = 0$

$x_2 = 5 \quad y_2 = -2$

$x_3 = -3 \quad y_3 = 2$

Now,

$$AM = BM$$

$$\sqrt{(x_1-x_2)^2+(y_1-y_2)^2} = \sqrt{(x_1-x_3)^2+(y_1-y_3)^2}$$

$$\sqrt{(x-5)^2+[0-(-2)]^2} = \sqrt{[x-(-3)]^2+(0-2)^2}$$

$$\sqrt{(x)^2+(5)^2-2(x)(5)+(12)^2} = \sqrt{(x)^2+(3)^2+2(x)(3)+(-2)^2}$$

$$\sqrt{x^2+25-10x+144} = \sqrt{x^2+9+6x+4}$$

Square on both side

$$x^2+25-10x+144 = x^2+9+6x+4$$

$$x^2-x^2-10x-6x = 9-25+4-144$$

$$0-16x = 9-29-144$$

$$-16x = -20-144$$

$$-16x = -164$$

BC

CD

$$\begin{array}{r} 121 \\ 25 \\ \hline 146 \end{array}$$

$BC = \sqrt{65}$

$CD = \sqrt{(x_4 - x_3)^2 + (y_4 - y_3)^2}$

$= \sqrt{(-1-4)^2 + [5-(-6)]^2}$

$= \sqrt{(-5)^2 + (5+6)^2}$

$= \sqrt{25 + (11)^2}$

$= \sqrt{25+121}$

$= \sqrt{146}$

$DA = \sqrt{(x_1 - x_4)^2 + (y_1 - y_4)^2}$

$= \sqrt{(-5-(-1))^2 + (7-5)^2}$

$= \sqrt{(-5+1)^2 + (2)^2}$

$= \sqrt{(4)^2 + (2)^2}$

$= \sqrt{16+4}$

$= \sqrt{20}$

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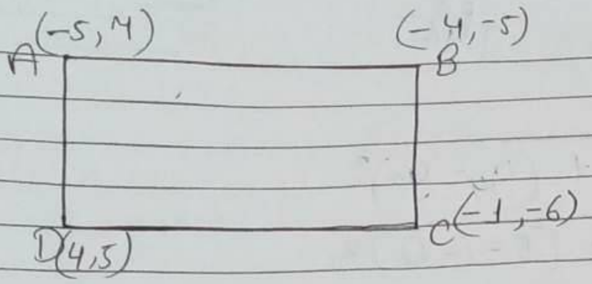
29.



$x_1 = -5$	$y_1 = 7$
$x_2 = -4$	$y_2 = -5$
$x_3 = 4$	$y_3 = -6$
$x_4 = -1$	$y_4 = 5$

Now,

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(-4-(-5))^2 + (-5-7)^2}$$
$$= \sqrt{(-4+5)^2 + (-12)^2}$$
$$= \sqrt{(-1)^2 + 144}$$
$$= \sqrt{1 + 144}$$
$$= \sqrt{145}$$

$$BC = \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}$$
$$= \sqrt{[4-(-4)]^2 + (-6-(-5))^2}$$
$$= \sqrt{(4+4)^2 + (-6+5)^2}$$
$$= \sqrt{(8)^2 + (-1)^2}$$
$$= \sqrt{64+1}$$

$\rightarrow \frac{2}{\sin\theta}$

$\rightarrow 2 \operatorname{cosec}\theta$

27.

28. Let the Tiru's age $= x$
& the age of Sagar $= y$

As per problem

Five years ago -

$2(x-5) = y-5$

$2x-10 = y-5$

$2x-y = -5+10$

$2x-y = 5$ — (1)

now, ten years later -

$y+10 = (x+10)+10$

$y+10 = x+10+10$

$y+10 = x+20$

$y-x = 20-10$

$y-x = 10$ — (2)

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25.

Given

In the given figure AB || DC.

To proof $\Delta AOB \sim \Delta DOC$

Prove:- In ΔAOB & ΔDOC

$\angle AOB = \angle COD$ [vertically opposite]

$\angle ABO = \angle OCD$ [Alternate angle]

$\angle BAO = \angle ODC$

By using AAA similarity

$\Delta AOB \sim \Delta DOC$

Section C

26. $$\frac{\cos(90^\circ-\theta)}{1+\sin(90^\circ-\theta)} + \frac{1+\sin(90^\circ-\theta)}{\cos(90^\circ-\theta)}$$

Sol⇒ $$\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}$$

$$\Rightarrow \frac{\sin^2\theta + (1+\cos^2\theta)}{(1+\cos\theta)\sin\theta}$$

$$\Rightarrow \frac{\sin^2\theta + (1)^2 + 1\cos^2\theta + 2(1)(\cos\theta)}{(1+\cos\theta)\sin\theta} \quad \left((a^2+b)^2 = a^2+b^2+2ab\right)$$

$$\Rightarrow \frac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{(1+\cos\theta)\sin\theta} \quad (\sin^2\theta+\cos^2\theta = 1)$$

$$\Rightarrow \frac{2+2\cos\theta}{(1+\cos\theta)\sin\theta} \Rightarrow \frac{2\cancel{(1+\cos\theta)}}{\cancel{(1+\cos\theta)}\sin\theta}$$

23.

We

Dis

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23.

Co-ordinates are -

$x_1 = 0 \qquad y_1 = 0$

$x_2 = -6 \qquad y_2 = 8$

We know that

$$\text{Distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$= \sqrt{(0-(-6))^2 + (0-8)^2}$$

$$= \sqrt{(0+6)^2 + (-8)^2}$$

$$= \sqrt{36+64}$$

$$= \sqrt{100}$$

$$= 10$$

Hence, The answer is 10.

24.

$$\cos(90° - 3A) = \cos(A - 10°)$$

$$90° - 3A = A - 10°$$

$$-3A - A = -10° - 90°$$

$$\cancel{+}4A = \cancel{+}100°$$

$$A = \frac{\cancel{100°}\,25}{\cancel{4}}$$

$$\boxed{A = 25°}$$

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22.

$4x + 6y = 7$; $kx + 18y = 21$

Compare this equation to $ax + by + c = 0$

so,

$a_1 = 4$, $a_2 = k$

$b_1 = 6$, $b_2 = 18$

$c_1 = 7$, $c_2 = 21$

now,

$$\frac{a_1}{a_2} = \frac{4}{k}$$

$$\frac{b_1}{b_2} = \frac{\cancel{6}\,1}{\cancel{18}\,3} = \frac{1}{3}$$

$$\frac{c_1}{c_2} = \frac{\cancel{7}\,1}{\cancel{21}\,3} = \frac{1}{3}$$

now,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{4}{k} = \frac{1}{3} = \frac{4}{3}$$

No, it does not have infinitely many solutions.

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16. Intersecting or coincident

17. Parallel

18. 16 : 81

19. Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explaination of A.

20. Both A and R are true and R is the correct explaination of A.

Section-B

21.

$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$\Rightarrow 30 \times 4 \times 3 \times 2 \times 1 + 5$

$\Rightarrow 30 \times 24 + 5$

$\Rightarrow 720$

Now, Factorisation

2	720
2	360
2	180
2	90
3	45
3	15
5	5
	1

$720 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5$

Here, we can see it has more than 2 factors that's why it is a composite number.

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14. $\cancel{x^5y^5z^6}$ $x^5y^5z^6$

15. $\frac{x^2}{2} - \frac{x}{2} - 6$